Subject: ENVIRONMENTAL IMPACT ASSESSMENT

Subject Code: 23A01504c

UNIT – I

Concepts and Methodologies of EIA

I. Introduction to EIA

Environmental Impact Assessment (EIA) is a systematic process of identifying, predicting, evaluating and mitigating the biophysical, social and other relevant effects of a proposed project BEFORE major decisions and commitments are made.

Definition EIA is a planning and decision-making tool. It tells us what the environmental consequences of a development project will be, so that harmful effects can be avoided, reduced or compensated.

Objectives of EIA

- To predict the environmental effects of a proposal early in the planning stage.
- To find ways and means to reduce the adverse effects.
- To shape the project to suit the local environment.
- To present the predictions and options to the decision makers.
- To promote sustainable development and public participation.

Need for EIA

- Rapid industrialisation, urbanisation and mining damage air, water, soil and biodiversity.
- Prevention is cheaper and better than cure (treatment after damage).
- Legal requirement in India under the EIA Notification 2006 (Environment Protection Act 1986).
- Gives transparency in decisions and builds public trust.

2. Initial Environmental Examination (IEE)

IEE is a preliminary, quick and low-cost study done at the early stage of a project to decide whether a full EIA is needed or not.

- **Purpose** screening of the project - to find out if significant impacts are likely.
- It uses existing data and expert judgement; no long field surveys.
- Output of IEE: (a) no further study needed, (b) mitigation measures are enough, or (c) detailed EIA is required.

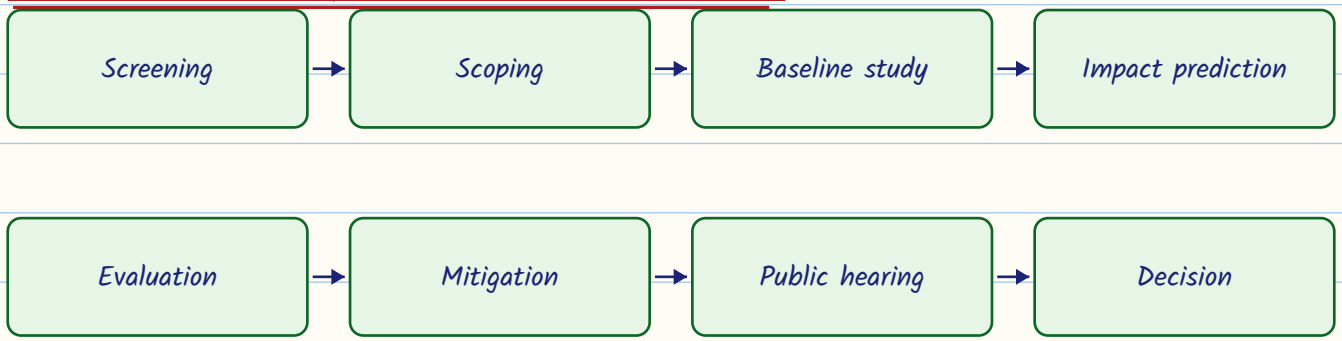
Contents of an IEE report

- Description of the project and its location.
- Description of the existing environment (baseline in brief).
- Likely impacts - positive and negative.
- Possible mitigation measures and alternatives.
- Conclusion and recommendation for full EIA.

IEE vs EIA

Point	IEE	EIA
Depth	Preliminary, brief	Detailed, comprehensive
Time and cost	Less	More
Data	Secondary data mostly	Primary field data and monitoring
Result	Decides need for EIA	Final basis for clearance

3. Elements (Steps) of the EIA Process



- **Screening** decides whether the project needs EIA and at what level (category A or B).
- **Scoping** identifies key issues, alternatives and terms of reference (ToR) for the study.
- **Baseline study** collection of data on the existing air, water, soil, noise, flora, fauna and socio-economic status.
- **Impact identification and prediction** finding what changes the project will cause and how big they are.
- **Impact evaluation** judging the significance of the impacts against standards and public values.
- **Mitigation** measures to avoid, minimise, restore or compensate impacts.
- **Environmental Management Plan (EMP)** who will do what, when and at what cost.
- **Public consultation** hearing the views of affected people.
- **Review and decision** expert committee appraises the report and grants or rejects clearance.
- **Monitoring and follow-up** checking that conditions are followed after the project starts.

4. Factors Affecting EIA (Impact Evaluation and Analysis)

The quality and result of an EIA depend on many factors.

- **Nature and size of the project** a thermal power plant has far more impact than a small workshop.
- **Location and sensitivity of site** near forests, rivers, wetlands, protected areas or dense population increases impact.
- **Quality of baseline data** wrong or old data gives wrong predictions.
- **Prediction methods used** models, experience, simulations.

- **Time scale** short-term, long-term, temporary or permanent impacts.
- **Space scale** local, regional, national or trans-boundary.
- **Reversibility** can the environment recover or not.
- **Public participation** concerns of local people change the final decision.
- **Legal standards and policies** permissible limits for air, water and noise.
- Expertise and funds available to the study team.
- **Alternatives available** other sites, technologies or designs.

Impact evaluation and analysis

Evaluation means judging the importance of a predicted impact. Common criteria are:

- Magnitude (how large) and extent (how wide an area).
- Duration and frequency of the impact.
- Probability (chance of occurring) and reversibility.
- Whether standards or legal limits are crossed.
- Value of the affected resource and number of people affected.
- Cumulative effect with other existing projects.

Impact types direct and indirect; positive and negative; short term and long term; reversible and irreversible; local and regional; cumulative.

5. Preparation of Environmental Base Map

An environmental base map is a map of the study area on which the existing environmental features, sensitive places and the proposed project are marked. It is the starting point for all further analysis.

Steps in preparation

- Select a suitable scale (for example 1:50,000 toposheet of Survey of India).
- Mark the project site and the study area boundary (commonly 10 km radius around the site).

- Show topography, contours, drainage, rivers, lakes and canals.
- Mark land use and land cover – agriculture, forest, settlements, industries.
- Mark roads, railways and other infrastructure.
- **Show sensitive receptors** schools, hospitals, temples, protected areas, wildlife corridors.
- Show sampling locations of air, water, soil and noise monitoring.
- Add north direction, legend, scale and wind rose.

Uses of the base map

- Helps in locating the project properly and in alternative site selection.
- Used for overlaying other maps in the overlay method.
- Helps in understanding spatial spread of impacts.
- Can be prepared using remote sensing and GIS for better accuracy.

6. Classification of Environmental Parameters

Environmental parameters are the measurable features of the environment that may be changed by a project. They are classified into four groups.

Category	Parameters (examples)
Physical / Chemical	Air quality, water quality, soil quality, noise, climate, geology, hydrology
Biological	Flora, fauna, aquatic life, forests, endangered species, habitats
Socio-economic	Population, employment, health, income, housing, transport, education
Cultural / Aesthetic	Historical sites, monuments, religious places, landscape beauty, recreation

- Each parameter is described by indicators, for example BOD and DO for water, SO2 and NOx for air, dB(A) for noise.
- Parameters are given importance weights in some methods (like EQI and

matrix).

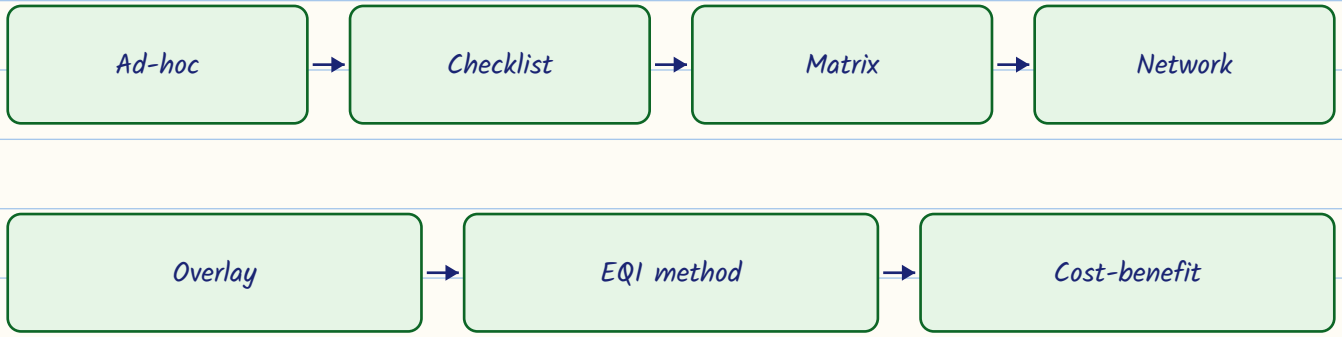
7. Criteria for Selection of EIA Methodology

No single method suits all projects. The choice depends on the following criteria.

- Type, size and complexity of the project.
- Type and quantity of data available.
- Stage of planning (early or detailed).
- Time, money and skilled manpower available.
- Ability of method to identify and compare alternatives.
- Ability to handle direct, indirect and cumulative impacts.
- Ability to give quantitative results.
- Whether it can be understood by the public and decision makers.
- Requirement of the law or the funding agency.
- Flexibility and reproducibility of the method.

8. EIA Methods

EIA methods are tools used to identify, predict and evaluate impacts. The main ones are:



8.1 Ad-hoc Methods

- These give minimum guidance; experts simply list the broad areas of impact based on their experience.

- Impacts are described in general terms - no fixed format.
- **Advantages** quick, cheap, useful for small projects and for a first look.
- **Disadvantages** no uniformity, depends on the experts, can miss important impacts, subjective.

8.2 Checklist Methods (related)

- A list of environmental factors to be checked against the project activities.
- **Types** simple (list only), descriptive (with notes on measurement and prediction), scaling (with ratings) and scaling-weighting checklists.
- Simple to use and good for screening, but do not show interaction among impacts.

8.3 Matrix Methods

A matrix shows project activities along one axis and environmental factors along the other. A mark is placed in a cell where an activity affects a factor.

Leopold Matrix (1971) developed by Luna Leopold of the US Geological Survey. It has 100 project actions (columns) and 88 environmental characteristics (rows), giving 8800 possible interactions.

- In each cell a slash is drawn when an impact is possible.
- Then two numbers from 1 to 10 are written: MAGNITUDE (top left, how big - with + or - sign) and IMPORTANCE (bottom right, how significant).
- Example cell: 8 over 5 means large magnitude, medium importance.
- **Advantages** covers many actions and factors, shows direct link between cause and effect, easy to present.
- **Disadvantages** does not show secondary (indirect) impacts, does not consider time and space, values are subjective, very large for big projects.

Project activity	Air	Water	Soil	Noise	Jobs
Land clearing	-	-	-	-	+
Construction	-	-	-	-	+

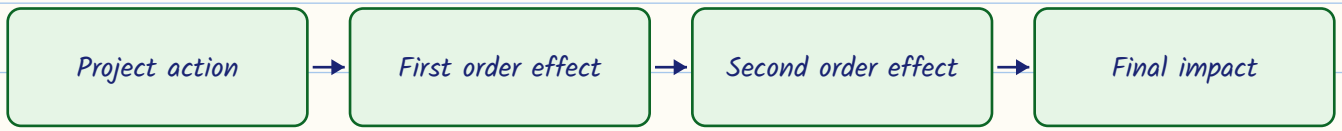
Operation	-	-	-	-	+

Other matrix types

- **Simple interaction matrix** only marks presence or absence of impact.
- **Weighted (quantitative) matrix** gives numerical values and totals them.
- **Stepped (cross-impact) matrix** tries to show second and third order effects.

9. Network Method

- Network (flow diagram) methods link the causes and effects as a chain, so that direct and indirect impacts can be seen together.
- Developed to overcome the drawback of matrices which show only first order effects.
- **Sorensen network (1971)** made for coastal zone planning. It starts from land use activities, then shows the causal factors, the environmental changes and finally the impacts on people.



- **Example** Deforestation for a road -> soil erosion -> siltation of river -> loss of fish -> loss of income of fishermen.
- **Advantages** shows indirect and cumulative impacts, good for understanding cause-effect pathways.
- **Disadvantages** becomes complex and untidy for large projects, mostly qualitative, no ranking of importance.

10. Environmental Media Quality Index Method

- Also called the Environmental Quality Index (EQI) or Battelle Environmental Evaluation System (EES) method, developed by Battelle Columbus Laboratories

(1972) for water resources projects.

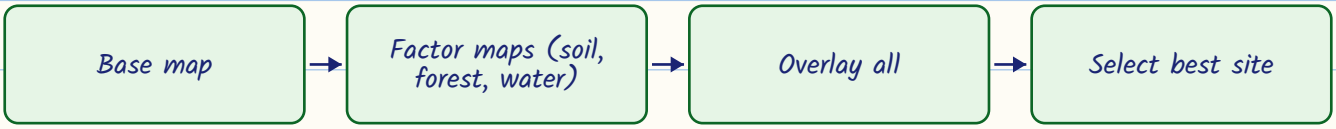
- **Environment is divided into 4 categories** Ecology, Environmental Pollution, Aesthetics and Human Interest.
- These have 18 components and 78 parameters.
- A total of 1000 Parameter Importance Units (PIU) are shared among all parameters (weights).
- For each parameter, the measured value is converted into an Environmental Quality (EQ) scale from 0 (very bad) to 1 (very good) using a value function curve.

Formula Environmental Impact Units (EIU) = sum of [PIU x EQ]. The index is calculated for "with project" and "without project" conditions; the difference is the net impact.

- **Advantages** gives numerical and comparable results, weights show relative importance, good for comparing alternatives.
- **Disadvantages** costly and time consuming, value functions are subjective, needs a lot of data.

II. Overlay Methods

- Overlay method uses a set of transparent maps (mylar sheets), one for each environmental factor.
- Each map is shaded darker where the sensitivity or impact is higher.
- When all the maps are placed over one another on the base map, the combined darkness shows areas of high total impact and low impact.



- **McHarg's overlay method (1969)** Ian McHarg used it for highway route

selection, choosing the path with the least total environmental cost.

- Today the same idea is done with GIS using digital layers (GIS overlay analysis).
- **Advantages** easy to understand, good for site and route selection, spatial picture of impact.
- **Disadvantages** all factors get equal importance unless weighted, cannot show indirect or time based impacts, limited number of sheets can be read clearly.

12. Cost / Benefit Analysis

- A method of comparing the total costs of a project with its total benefits, both expressed in money terms.
- Includes environmental costs (loss of forest, health damage) and environmental benefits (clean energy, flood control) along with economic ones.
- **Benefit Cost Ratio** BCR = Present value of benefits / Present value of costs. A project is acceptable if BCR is greater than 1.
- **Net Present Value** NPV = PV of benefits - PV of costs; must be positive.
- Future values are brought to present using a discount rate: $PV = F / (1 + r)$ raised to n.

Steps

- Define the project and alternatives.
- Identify all costs and benefits, direct and indirect.
- Give money values (market price, replacement cost, willingness to pay).
- Discount to present value and compare.
- Do sensitivity analysis for different discount rates.
- **Advantages** simple decision rule, shows economic efficiency.
- **Disadvantages** difficult to put a price on life, health, beauty and species; discount rate favours present generation; ignores who gains and who loses.

13. Comparison of Methods (Quick Revision)

Method	Key feature	Main limitation
Ad-hoc	Expert opinion, quick	Subjective
Checklist	List of factors	No interaction shown
Matrix	Action vs factor grid	No indirect impact
Network	Cause-effect chain	Complex, qualitative
EQI / Battelle	Numerical weighted index	Costly, data heavy
Overlay	Map layers	Equal weight, no time
Cost-benefit	Money terms	Pricing of environment

Important questions (1) Define EIA and explain its elements. (2) Differentiate IEE and EIA. (3) Explain the Leopold matrix. (4) Describe the EQI method. (5) Write short notes on overlay and network methods. (6) Criteria for selecting an EIA method.

14. Worked Illustration – Leopold Matrix

Suppose a cement factory is proposed near a village. Land clearing is Action A, Quarrying is Action B and Stack emission is Action C. Environmental factors are air quality, groundwater and employment.

Factor	A Clearing	B Quarrying	C Stack emission
Air quality	-3 / 4	-6 / 7	-9 / 9
Groundwater	-2 / 3	-7 / 8	-1 / 2
Employment	+5 / 6	+7 / 7	+4 / 5

- Each entry is Magnitude / Importance. A minus sign shows adverse impact and a plus sign shows beneficial impact.
- Add the cells row-wise to see which factor is hit most and column-wise to see which action is the worst.
- Here stack emission is the most harmful for air quality; so the main mitigation

is an air pollution control system such as an ESP and a tall stack.

Remember the matrix gives only a first look. It must be followed by detailed prediction using models and field data.

15. Environmental Management Plan (EMP) in Brief

- EMP is the document that turns EIA findings into action.
- It lists mitigation measures, responsible persons, schedule, budget and monitoring programme.
- **Parts** air pollution control plan, water and effluent plan, solid waste plan, green belt plan, noise control, occupational health and safety, disaster management and rehabilitation plan.
- A green belt around the project reduces dust and noise and improves appearance.
- Regular monitoring reports are submitted to the pollution control board and the ministry.

16. Key Points to Remember

- EIA is preventive; it is done before the project starts.
- IEE is the quick study that decides whether full EIA is required.
- Baseline data are collected for at least one season, normally three months for air and water.
- Ad-hoc is the simplest method; EQI is the most quantitative; overlay is best for spatial planning; network is best for indirect impacts.
- Leopold matrix = 100 actions x 88 factors; Battelle system = 1000 PIU, 78 parameters.
- Cost-benefit analysis is acceptable when BCR > 1.